

ClubHouse

Multi-College Student Collaboration Platform

FULL TECHNICAL DOCUMENTATION v2 — Refreshed with all improvements + live deployment

Item	Detail
Workspace	D:\OpenCodeProjs\Clubhouse\Web-Mobile-Fit — branch main on github.com/harsha-2oo8/-ClubHouse
Date / version	20 September 2026 — v2 (v1 was pre-improvement audit)
Stack	React 19 + Vite 7 + Express 5 + PostgreSQL 16 + Drizzle + Clerk + Tailwind v4 + pnpm
Production	LIVE: Vercel SPA + Render API (clubhouse-api-yxsm) + Supabase Postgres (ap-northeast-2), 18 tables verified
API contract	OpenAPI 3.1 — 48 paths (added /readyz) + limit params, codegen in sync
Database	18 tables incl. new admin_audit_logs; unique constraints + indexes added
Auth	Clerk + Bearer bridge (ClerkTokenBridge) for cross-origin + env admin bootstrap
Quality gates	pnpm typecheck green (all pkgs), 11 Vitest tests pass, CI with codegen-drift check

Generated from the live repository state after the full improvement program (Phases 0–4, 6–9 part, 13–16 part) and the production deploy.
New/changed since v1: RBAC, bootstrap, constraints, storage auth, rate limits, audit log, pagination, tests, CI, .env.example, 5 docs, render.yaml/vercel.json, VITE_API_URL wiring, Bearer fix, migration snapshot, live topology.

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1. Executive summary & production status

ClubHouse is a multi-college student collaboration platform: 2-step onboarding, colleges + membership + moderator applications, clubs, projects (apply/invite/members/chat/calendar), events + registration, notifications, admin approvals, file uploads. Since v1 it went from Replit prototype to **live production**: Vercel SPA → Render API → Supabase Postgres, with a security/integrity hardening pass (env admin bootstrap, centralized RBAC, DB constraints, storage auth, rate limits, audit logging, pagination, tests, CI). Verified: typecheck green everywhere, 11/11 tests pass, all 3 packages build, codegen in sync, 18/18 tables live in Supabase, **/api/readyz ok**.

2. Monorepo & workspace (new files marked *)

Package	Entry	Notes
artifacts/api-server	src/index.ts → app.ts	Express 5, /api, esbuild ESM dist/index.mjs
artifacts/clubhouse	src/main.tsx → App.tsx	React 19 SPA, dist/ (Vercel), VITE_API_URL wiring
artifacts/mockup-sandbox	UI sandbox	Not shipped
lib/db	schema + drizzle/ snapshot*	push workflow; *0000_strong_sentinel.sql + meta committed
lib/api-spec	openapi.yaml	HTTP truth; added /readyz + limit params
lib/api-client-react/zod	generated/*	Orval output — never hand-edit
lib/object-storage-web	Uppy uploader	Works on Replit; needs GCS key off-Replit
api-server/src/lib/*	*params, adminBootstrap, policy, notify, rateLimit, audit	New cross-cutting modules + 2 test files
*.yaml/json (root)	*render.yaml, vercel.json, railway.json, .env.example	Deploy blueprints + env template
.github/workflows/ci.yml*	CI	install → typecheck → tests → codegen-drift → build
docs/*	*5 markdown docs	IMPROVEMENT_PLAN, ARCHITECTURE, SECURITY, DEVELOPMENT, TESTING

3. Technology stack

Layer	Choice	Notes
Language/runtime	TypeScript 5.9, Node 24	strict; pnpm 10 workspaces + catalog
Frontend	React 19.1, Vite 7, wouter, TanStack Query 5	Data only via generated Orval hooks
UI	shadcn/new-york, Radix, Tailwind v4, next-themes	Indigo primary 245 80% 58%
Backend	Express 5.2, pino, cors	clerkMiddleware; PORT env (Render provides)
DB	Supabase Postgres 16 + Drizzle + pg	Session pooler :5432 (IPv4); direct host is IPv6-only
Auth	Clerk	Same pk_... web+API; sk_... API only; Bearer bridge cross-origin
Validation/codegen	Zod, Orval 8.18, drizzle-zod	Spec-first; CI fails on drift
Tests/CI	Vitest 3.2, GitHub Actions	11 unit tests; Supertest matrix open

4. Frontend (artifacts/clubhouse)

Providers: WouterRouter (base), QueryClient (retry 1, 30 s stale), ThemeProvider, TooltipProvider, Toaster, ClerkProvider + **ClerkTokenBridge**. **main.tsx** reads **VITE_API_URL** and calls **setBaseUrl** so all calls go to Render. **ClerkTokenBridge (App.tsx)** registers **setAuthTokenGetter()** => **getToken()** so every request carries **Authorization: Bearer** — required because Vercel→Render is cross-origin and cookies never travel. Also fixed since v1: Orval v8 hook arity (undefined params), unknown→ReactNode guards, event visibility **college_only** to match spec.

Routes (18, wouter)

Route	Page	Purpose
/	Landing	Public marketing
/sign-in/*, /sign-up/*	Clerk	Fallbacks /dashboard, /onboarding
/onboarding	Onboarding	2-step profile → PATCH /users/me
/dashboard	Dashboard	Stats + activity
/discover	Discover	Projects search/filter + create
/discover/colleges	DiscoverColleges	Colleges + join + register
/discover/events	DiscoverEvents	Events + register + create
/clubs, /clubs/register, /clubs/:id, /clubs/:id/admin	clubs.tsx	Directory/register/profile/owner-admin
/colleges/:id	CollegePage	Overview/members/projects/meetings
/projects/:id	ProjectPage (largest)	Chat (REST) + calendar + invites + applications
/profile/me, /profile/:id	ProfilePage	Self-editable
/notifications	NotificationsPage	Grouped, mark-read
/admin	AdminPage	Approvals + stats (admin only)

5. Backend (artifacts/api-server)

Pipeline: pino-http → /api/__clerk proxy → CORS (reflect) → JSON → clerkMiddleware → /api router. New modules: **params.ts** (getParam/getIntParam/getLimit — fixes Express 5 string|string[] typing), **adminBootstrap.ts**, **policy.ts** (isAdmin/isCollegeMember/isCollegeModerator/isProjectMember/isProjectOwner/isClubOwner, can* helpers, require* middleware), **notify.ts** (notificationService), **rateLimit.ts** (20 writes/min, 60 searches/min, 429+Retry-After, on all write routes + search + uploads), **audit.ts** (recordAuditLog, secret-scrubbed). New route: **GET /readyz** (SELECT 1 DB probe). Storage: private objects require session + ACL + traversal guards; upload minting authed + limited.

Routes (~65 ops, all under /api)

Domain	Highlights
health	GET /healthz (liveness), GET /readyz* (readiness)
users	GET/PATCH /me (bootstrap), GET /search?limit (rate-limited)
colleges	CRUD + join lifecycle + moderator applications + meetings; ?limit on list
clubs	CRUD + members + management events; POST requires an approved college
projects	CRUD + invites/apply/applications + messages + events; ?limit on list
events	Catalog + registration (unique); ?limit on list
notifications/dashboard/admin	Feed + stats + approvals (requireAdmin, audited, limited)
storage	Presigned PUT (authed/limited); private serve (authed+ACL)

6. Database — 18 tables

Supabase Postgres, Drizzle truth in lib/db/src/schema + committed snapshot **lib/db/drizzle/0000_strong_sentinel.sql** (enables incremental generates; push introspection is unreliable against Supabase). Added: **admin_audit_logs**; unique membership/registration + status-scoped application uniqueness; cascade deletes; indexes on hot FKs, message chronology, notifications; **club_members.clerk_id** nullable link; users.college kept display-only; required_roles/partner_colleges stay validated JSONB by decision.

Tables (18)	Constraints/indexes
users	email unique; PK clerk_id
colleges, college_members, college_join_requests, moderator_applications, college_meetings	unique member; unique request/app scoped by status; cascade; member/meeting indexes
projects, project_members, project_applications, project_messages, project_events	unique member; status-scoped applications; (project,created)/(project,id) msg indexes
clubs, club_members, club_management_events	club/clerk indexes; clerk_id link
club_events, event_registrations	unique (event,clerk)
notifications, admin_audit_logs*	clerk/created + clerk/read indexes; audit admin/entity/created indexes

7. API contract

openapi.yaml: added **GET /readyz**, optional clamped **limit** on listProjects/listColleges/listEvents/searchUsers (arrays unchanged — backward compatible). Regenerated clients in sync (verified by double codegen + CI drift check).

8. Auth & RBAC

Clerk only, no OTP. Web: ClerkProvider + VITE_API_URL + **Bearer bridge**. API: clerkMiddleware; **policy.ts** enforces membership/ownership from DB on every protected route (UI gating is cosmetic). Admin bootstrap: **ADMIN_EMAILS/ADMIN_CLERK_IDS** grant **role=admin only at first profile creation**; afterwards DB-controlled; remove vars post-bootstrap. Hardcoded admin email fully removed (grep-verified).

9. Storage

GCS via presigned PUT (900 s). Private serve requires Clerk session + ACL metadata + rejects '..'. **Production caveat: minting uses the Replit sidecar — absent on Render — so logo/brochure uploads fail in prod until migrated to a GCS service-account key.** Clerk avatars unaffected. Public-asset path unchanged.

10. Config, env, commands

Variable	Scope	Notes
DATABASE_URL	API (Render)	Supabase Session pooler :5432 verbatim; direct db.* host is IPv6-only
CLERK_PUBLISHABLE_KEY	API	Backend verification
VITE_CLERK_PUBLISHABLE_KEY	Web build	Baked at build time
VITE_API_URL	Web build	Render origin, no trailing slash; build-time
CLERK_SECRET_KEY	API (prod)	Token verification + proxy; never frontend
ADMIN_EMAILS/IDS	API (temp)	First-boot only; remove after
PORT / BASE_PATH	API / legacy	PORT from Render; web no longer requires either

Commands: pnpm install (--ignore-scripts on Windows) · pnpm run typecheck · pnpm run build (Vite pkgs need PORT/BASE_PATH only on legacy config) · api-spec codegen · db push (dev) / generate+apply (Supabase) · api-server test.

11. Build, typecheck, CI, deploy

Typecheck green (libs + api-server + clubhouse + sandbox + scripts). Tests 11/11. Builds: api esbuild bundle, clubhouse+sandbox Vite (needs env on legacy config). CI (.github/workflows/ci.yml): install → typecheck → tests → **codegen-drift fail** → build. Deploy files: **vercel.json** (build output artifacts/clubhouse/dist + SPA rewrite), **render.yaml** (Node 24, health /api/healthz), railway.json (legacy).

12. Production topology & deploy lessons

Layer	Target	Verified
DB	Supabase ap-northeast-2, Session pooler	18/18 tables queried back OK
API	Render clubhouse-api-yxsm (free)	/api/healthz ok, /api/readyz ok after pooler URL fix
Web	Vercel (vercel.json)	Bearer fix auto-deployed from main
Auth	Clerk existing keys	401 storm resolved by token bridge

Lessons (bought with real errors): (1) split-origin SPA must send Bearer tokens — cookies never cross Vercel→Render; (2) Supabase direct host is IPv6-only — Render + many dev machines need the Session pooler; (3) passwords with [] break URI parsing — percent-encode or strip; (4) drizzle-kit push introspection fails silently on Supabase — use committed generate+apply; (5) drizzle-kit schema paths must be POSIX on Windows.

13. User flows & roles

Signup → onboarding → dashboard → discover/join colleges → create/apply/invite projects → chat/events → clubs → event registration → notifications → admin approvals → profile edits — all verified live except file uploads (pending GCS migration). Roles: guest (browse) < member < moderator (own college) < club owner < platform admin (requireAdmin).

14. Decisions, gotchas & lessons

Kept: JSONB roles/partners (validated, not normalized); users.college display-only; array responses with clamped limits (no envelope churn); pnpm supply-chain minimumReleaseAge. Gotchas: Express 5 params via getParam; never id in insert omits; Orval arity (undefined params); Boolean() for unknown JSX; VITE_* need redeploy; Render free sleeps (~60 s cold start); Supabase free pauses after a week idle (one-click resume).

15. Tests & roadmap

Vitest: params (6) + adminBootstrap (5) = 11 green. Next: Supertest authz matrix (10 priority cases), cursor envelopes, SSE chat + polling fallback, trigram search facets, ProjectPage split, bottom-nav mobile, GCS service-account uploads, audit-log UI. Deferred deliberately — recorded in docs/IMPROVEMENT_PLAN.md §7.

Appendix — tree & cheat sheet

```
Root: vercel.json render.yaml railway.json .env.example package.json pnpm-workspace.yaml · docs/ (5)
· .github/workflows/ci.yml · artifacts/{api-server, clubhouse, mockup-sandbox} · lib/{db
(+drizzle/), api-spec, api-client-react, api-zod, object-storage-web} · scripts/ · *.pdf guides.
```

```
Cheat: dev API `pnpm --filter @workspace/api-server run dev` · web `pnpm --filter
@workspace/clubhouse run dev` · typecheck `pnpm run typecheck` · tests `pnpm --filter
@workspace/api-server run test` · codegen `pnpm --filter @workspace/api-spec run codegen` · db `pnpm
--filter @workspace/db run push` (local) · prod: readyz → Vercel redeploy on push to main.
```

— End of v2 — Repo: github.com/harsha-2008/ClubHouse (main) — Live: Vercel + Render + Supabase —